



Barker
College

2024

YEAR 11
PRELIMINARY EXAMINATION

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Student Number

Mathematics Advanced

Staff Involved:

- DZB • SJB • JZT • DZP • ALY • ARP
- ECB* • FHD • RJW • DXC • AYH • BHC

Tuesday 10 September 2024

210 copies

TOPICS COVERED:		
Algebra, Number and Surds	Functions and Graphs	Transformations and Symmetry
The Coordinate Plane	Trigonometry	Exponential and Logarithmic Functions
Differentiation and Curve Sketching	Extending Calculus	Probability

General

Instructions:

- Reading time - 5 minutes
- Working time - 2 hours 30 minutes
- Write your Student Number on all answer pages
- Calculators approved by NESA may be used
- A separate reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and / or calculations

Total

marks:

100

Section I - 10 marks (pages 2 - 5)

- Attempt Questions 1 - 10
- Allow about 15 minutes for this section

Section II - 90 marks (pages 9 - 28)

- Attempt Questions 11 - 20
- Show all necessary working
- Allow about 2 hours and 15 minutes for this section

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple choice answer sheet for Questions 1 – 10.

- 1 Determine the gradient of the line perpendicular to $y = \frac{x}{2} - 5$.

- A. -2
- B. $-\frac{1}{2}$
- C. $\frac{1}{2}$
- D. 2

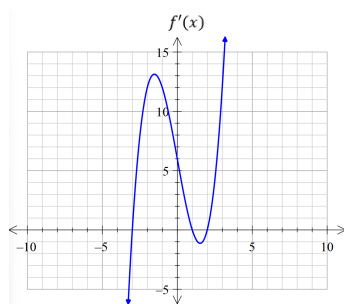
- 2 What is the value of $\log_{31} 586$ correct to **three significant figures**?

- A. 1.856
- B. 1.86
- C. 2.768
- D. 2.77

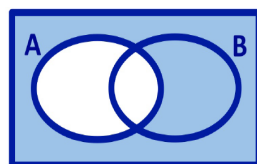
- 3 Given that $\log_a 4 = 0.420$ and $\log_a 5 = 0.488$. What is the value of $\log_a 100$?

- A. 0.410
- B. 0.664
- C. 1.396
- D. 2.908

- 4 The graph of the gradient function, $f'(x)$, is shown below. For what value(s) of x does the function, $f(x)$, have stationary points?



- A. $x = -1.5$ and $x = 1.5$
 B. $x = -3$, $x = 1$ and $x = 2$
 C. $x = -3$, $x = -1.5$, $x = 1$, $x = 1.5$ and $x = 2$
 D. $x = 1.5$ only
- 5 Determine the set notation represented by the **shaded area** in the Venn diagram below.



- A. B'
 B. A'
 C. $A' \cap B$
 D. $B \cup (A' \cap B')$

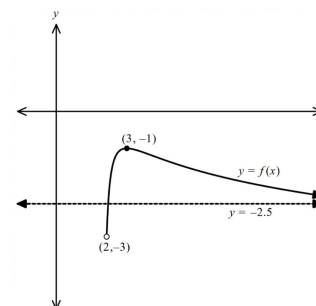
- 6 Determine the value(s) of k for which $|2k - 1| = k$

- A. 1 only
 B. $-\frac{1}{3}$ only
 C. -1 or $-\frac{1}{3}$
 D. 1 or $\frac{1}{3}$

- 7 Determine the value of p so that $\frac{a^2 a^{-3}}{\sqrt{a}} = a^p$.

- A. -3
 B. 12
 C. $-\frac{1}{2}$
 D. $-\frac{3}{2}$

- 8 The graph of a function $y = f(x)$ is shown below. It has an asymptote at $y = -2.5$. Which statement **correctly** describes its domain and range?



- A. Domain: $(2, \infty)$; Range $[-2.5, -1]$
 B. Domain: $(-3, -1]$; Range $(2, \infty)$
 C. Domain: $[2, \infty)$; Range $[-3, -1]$
 D. Domain: $(2, \infty)$; Range $(-3, -1]$

- 9 What is the **area**, in cm^2 , of the sector with radius 8 cm and arc length $20\pi\text{ cm}$?
- A. 80π
B. 80
C. 160π
D. 20π
- 10 Given that f and g are two functions such that $f(x) = 2x$ and $g(x + 2) = 3x + 1$, which function is $f(g(x))$?
- A. $6x - 5$
B. $6x + 1$
C. $6x - 10$
D. $6x + 2$

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End of Section I

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Student Number



Barker
College

Mathematics Advanced

Section II Answer Booklet

90 marks

Attempt Questions 11 - 20

Allow about 2 hours and 15 minutes for this section

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Instructions:

- Write your Student Number at the top of the page where indicated.
 - Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
 - Your responses should include relevant mathematical reasoning and/or calculations.
 - Extra writing booklets are available.
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Student Number

Question 11 (9 marks)

- (a) **Solve** this pair of simultaneous equations:

$$\begin{aligned} 3x + y &= 6 \\ 2x - 3y &= 15 \end{aligned}$$

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- (b) **Solve** $3 - 5x \leq 2$

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- (c) If $f(x) = x^2 - 2x + 5$, find $\frac{1}{2}(f(-2) - f(2))$

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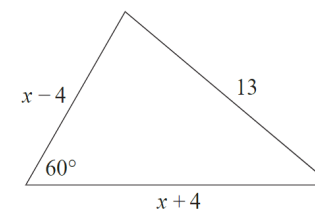
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- (d) Calculate the value of x in the diagram below.

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Student Number

Question 12 (9 marks)

- (a) Simplify $3^{2m} \times 3^{m-2} \times 3^m$ 1

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- (b) State the **domain** of the function $f(x) = \sqrt{x}$ 1

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- (c) Determine the equation of the line that is **parallel** to the interval joining the points $A(3, 5)$ and $B(6, 10)$, and passes through the point $C(-3, -1)$. 3
Leave your answer in **general form**.

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- (d) **Prove** whether $f(x) = \frac{x^4-1}{x}$ is even, odd or neither. 2

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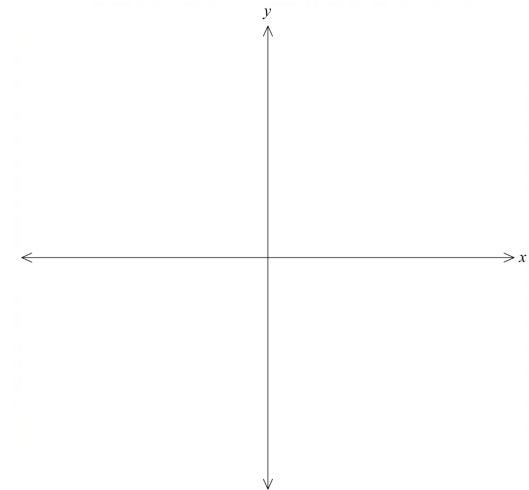
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- (e) Sketch the graph of $y - 2 = -2e^x$, **showing all intercept(s) and asymptote(s)**. 2



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Student Number

Question 13 (9 marks)

- (a) By first expressing in **index form**, solve $\log_5(2x + 1) = 2$, for x .

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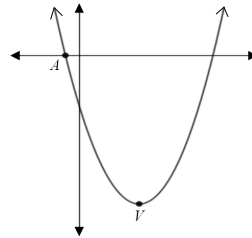
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- (b) The diagram shows the graph of $y = x^2 - 4x - 2$.



- i) Find the coordinate of the vertex, V .

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- ii) Find the x -coordinate of the intercept marked A , in **exact form**.

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- (c) The number of hours H that is take for a block of ice to melt varies inversely with the temperature T . At 32°C it takes 6 hours for a block of ice to melt.

- i) Find an **equation** for the graph in terms of H and T in the form $H = \frac{k}{T}$, where k is a constant.

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- ii) How much longer will it take the same size block of ice to melt at 10°C , compared to 25°C ? Leave your answer to the **nearest minute**.

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Student Number

Question 14 (9 marks)

- (a) Use the method of **first principles** to differentiate $f(x) = x^2 + 4x$, using: **3**

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

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- (b) Find the quadratic function $g(x)$ such that $g(0) = 16$ and $g(4) = g(-2) = 0$. **3**

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- (c) A packet of lollies contains 5 red lollies and 14 green lollies. Two lollies are selected at random **without replacement**. **2**

- i) Draw a **probability tree diagram** to illustrate the possible outcomes. **2**

- ii) What is the probability that the two lollies are **different colours**? **1**

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Student Number

Question 15 (9 marks)

(a) Differentiate and **simplify** the following functions with respect to x :

i) $f(x) = 2x^5 + 3x^2 - x + 1$

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ii) $f(x) = x^3 e^{2x}$

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iii) $f(x) = \frac{5x^2}{x^2+1}$

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iv) $f(x) = 3\sqrt{x} - \frac{4}{x^7}$

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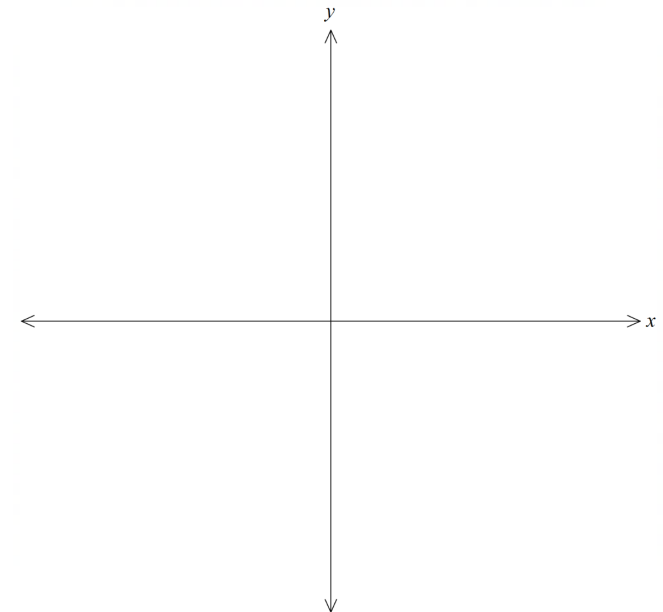
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(b) Sketch the graph of $f(x) = -(x-1)^2(x+2)(x+5)$, showing **ONLY** the x and y intercepts.

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Student Number

Question 16 (9 marks)

(a) Determine the solution(s) for the following equations in the interval $0 \leq \theta \leq 2\pi$

i) $2 \sin \theta = -\sqrt{3}$ **3**

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ii) $2 \cos^2 \theta - \cos \theta = 0$ **4**

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(b) **Prove** that $\sec x - \cos x = \sin x \tan x$

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Student Number

Question 17 (9 marks)

(a) A circle has equation $x^2 - 6x + y^2 + 4y + 9 = 0$

i) Determine the **centre** and **radius** of the circle.

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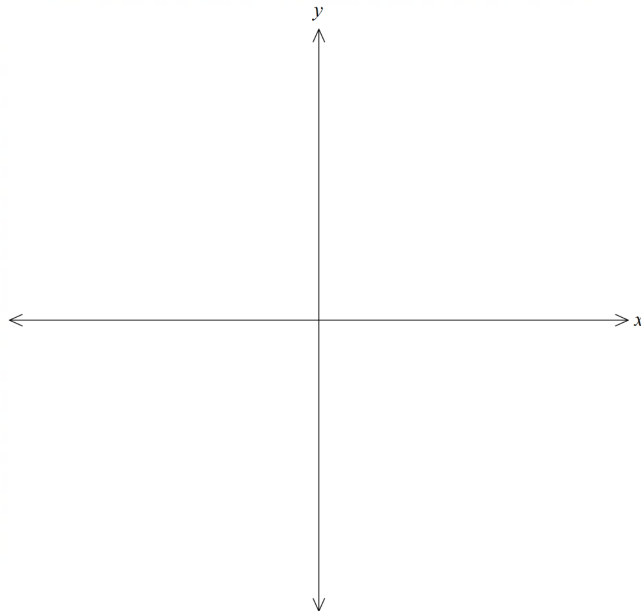
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ii) The circle is reflected across the x axis. **Sketch the reflected circle**, showing the coordinates of the **centre** and the **radius**.

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(b) Find the **equation** of the tangent to the curve $y = (2x + 1)^4$ at the point where $x = -1$.

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(c) A survey of the reading habits of some students revealed that, 25% read quality newspapers, 45% read tabloid newspapers and 40% do not read newspapers at all.

A student is selected at random. Given that this student reads newspapers, find the probability that this student **only** reads quality newspapers.

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Student Number

(a) The population of native penguins on an island has been studied since 1950 for conservation purposes.
The number of penguins, P , can be modelled by the equation $P = 2500e^{-0.036t}$, where t is the number of years after 1950.

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- [illegible]

- [illegible]

- [illegible]

- [illegible]

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Student Number

Question 19 (9 marks)

- (a) Determine the value(s) of m for which the equation $3x^2 - 2mx = -3$ has **one real solution**. **2**

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- (b) Solve for x in the following: $\log_4(4x + 16) - \log_4(x^2 - 2) = 1$ **3**

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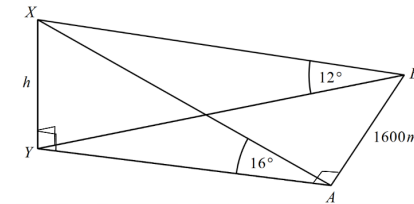
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- (c) Hunter walks 1600 m due North along a road from point A to point B . The point A is due east of a hill XY , where X is the top of the hill. The point Y is directly below point X and is on the same horizontal plane as the road. Let the height of the hill above point Y be h metres. From point A , the angle of elevation to the top of the hill is 16° . From point B , the angle of elevation to the top of the hill is 12° .



- i) Show that $BY = h \cot 12^\circ$ **1**

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- ii) Hence, find the value of h , correct to 2 decimal places. **3**

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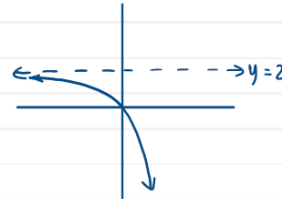
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1	$y = \frac{x}{2} - 5$ $= \frac{1}{2}x - 5$ $m = \frac{1}{2}$ $\perp m = -2$ (A)	8	$D: (2, \infty)$ $R: (-3, -1]$ (D)
2	$\log_{31}(586) = 1.86$ (B)	9	$l = r\theta$ $20\pi = 8\theta$ $\theta = \frac{20\pi}{8} = \frac{5\pi}{2}$ $A = \frac{1}{2}r^2\theta$ $= \frac{1}{2}(8)^2\left(\frac{5\pi}{2}\right)$ $= 80\pi \text{ cm}^2$ (A)
3	$\log_a(100) = \log_a(5^2 \times 4)$ $= \log_a(5^2) + \log_a(4)$ $= 2\log_a(5) + \log_a(4)$ $= 2(0.488) + (0.420)$ $= 1.396$ (C)	10	$g(x+2) = 3x+1$ $g((x+2)-2) = 3(x-2)+1$ $g(x) = 3x-6+1$ $= 3x-5$ $f(g(x)) = 2(3x-5)$ $= 6x-10$ (C)
4	Stat pts. occur when $f'(x)=0$ $\therefore$ at $x=-3, x=1$ and $x=2$ (B)		
5	A' (B)		
6	$ 2k-1 =k$ $2k-1=k$ or $2k-1=-k$ $k=1$ or $3k=1$ $k=\frac{1}{3}$ $\therefore k=1$ or $k=\frac{1}{3}$ (D)	11	(a) $3x+y=6$ $\times 3 \rightarrow 2x-3y=15$ (1) $9x+3y=18$ (2) $(1)+(2)$ $11x=33$ $x=3$ Sub $x=3$ into (2) $2(3)-3y=15$ $-3y=9$ $y=-3$ $\therefore x=3$ and $y=-3$
7	$\frac{a^2 a^{-3}}{\sqrt{a}} = \frac{a^{2+(-3)}}{a^{\frac{1}{2}}}$ $= \frac{a^{-1}}{a^{\frac{1}{2}}}$ $= a^{-1-\frac{1}{2}}$ $= a^{-\frac{3}{2}}$ $\therefore p = -\frac{3}{2}$ (D)		

(b)	$3-5x \leq 2$ $-5x \leq -1$ $x \geq \frac{1}{5}$	(d)	$f(x) = \frac{x^4-1}{x}$ $f(-x) = \frac{(-x)^4-1}{(-x)}$ $= \frac{x^4-1}{-x}$ $= -\frac{x^4-1}{x}$ $= -f(x)$ $\therefore$ ODD
(c)	$f(-2) = (-2)^2 - 2(-2) + 5$ $= 13$ $f(2) = (2)^2 - 2(2) + 5$ $= 5$ $\frac{1}{2}(f(-2) - f(2)) = \frac{1}{2}(13-5)$ $= 4$		(e) $y-2 = -2e^x$ $y = -2e^x + 2$ $\text{At } x=0$ $y = -2e^0 + 2$ $= 0$ $\therefore$ passes through $(0,0)$
(d)	$13^2 = (x-4)^2 + (x+4)^2 - 2(x-4)(x+4)\cos(60)$ $169 = x^2 - 8x + 16 + x^2 + 8x + 16 - 2(x^2 - 16)(\frac{1}{2})$ $= 2x^2 + 32 - x^2 + 16$ $= x^2 + 48$ $x^2 = 121$ $x = \pm 11$ However $x-4 > 0 \Rightarrow x > 4$ $\therefore x=11$		
12	(a) $3^{2m+m-2+m}$ $= 3^{4m-2}$ (b) $x \geq 0$ (c) $M_{AB} = \frac{10-5}{6-3}$ $= \frac{5}{3}$ Passes through $C(-3,-1), m = \frac{5}{3}$ $y - (-1) = \frac{5}{3}(x - (-3))$ $y + 1 = \frac{5}{3}(x + 3)$ $3y + 3 = 5x + 15$ $5x - 3y + 12 = 0$	13	(a) $\log_5(2x+1) = 2$ $2x+1 = 5^2$ $2x+1 = 25$ $2x = 24$ $x = 12$

(b) (i) $x = -\frac{b}{2a}$
 $= -\frac{(-4)}{2(1)}$
 $= 2$
 At $x=2$
 $y = (2)^2 - 4(2) - 2$
 $= -6$
 $\therefore V(2, -6)$
 (ii) $x = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(-2)}}{2(1)}$
 $= 2 \pm \sqrt{6}$
 $\therefore$ for A, $x = 2 - \sqrt{6}$

(c) (i) $H = \frac{k}{T}$
 At $T = 32$, $H = 6$
 $6 = \frac{k}{32}$
 $k = 192$
 $\therefore H = \frac{192}{T}$
 (ii) At $T = 10$
 $H = \frac{192}{10} = 19.2$
 At $T = 25$
 $H = \frac{192}{25} = 7.68$
 Difference $= 19.2 - 7.68$
 $= 11.52$
 $= 11 \text{ hours } 31 \text{ min}$

(b) $g(x) = ax^2 + bx + c$
 $g(0) = 16$
 $16 = a(0)^2 + b(0) + c$
 $c = 16$
 $g(4) = 0$
 $0 = a(4)^2 + b(4) + 16$
 $0 = 16a + 4b + 16$
 $0 = 4a + b + 4$ ①
 $g(-2) = 0$
 $0 = a(-2)^2 + b(-2) + 16$
 $0 = 4a - 2b + 16$
 $0 = 2a - b + 8$ ②
 ① + ②
 $0 = 6a + 12$
 $a = -2$
 Sub $a = -2$ into ①
 $0 = 4(-2) + b + 4$
 $b = 4$
 $\therefore g(x) = -2x^2 + 4x + 16$

(c) (i)

(ii) P(different colours)
 $= P(RG) + P(GR)$
 $= \left(\frac{5}{19} \times \frac{4}{18}\right) + \left(\frac{14}{19} \times \frac{5}{18}\right)$
 $= \frac{70}{171}$

14 (a) $f(x) = x^2 + 4x$
 $f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 + 4(x+h) - (x^2 + 4x)}{h}$
 $= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 + 4x + 4h - x^2 - 4x}{h}$
 $= \lim_{h \rightarrow 0} \frac{2xh + h^2 + 4h}{h}$
 $= \lim_{h \rightarrow 0} (2x + h + 4)$
 $= 2x + (0) + 4$
 $= 2x + 4$

15 (a) (i) $f'(x) = 10x^4 + 6x - 1$
 (ii) $u = x^3$ $v = e^{2x}$
 $u' = 3x^2$ $v' = 2e^{2x}$
 $f'(x) = uv' + vu'$
 $= (x^3)(2e^{2x}) + (e^{2x})(3x^2)$
 $= 2x^3e^{2x} + 3x^2e^{2x}$
 (iii) $u = 5x^2$ $v = x^2 + 1$
 $u' = 10x$ $v' = 2x$
 $f'(x) = \frac{vu' - uv'}{v^2}$
 $= \frac{(x^2+1)(10x) - (5x^2)(2x)}{(x^2+1)^2}$
 $= \frac{10x^3 + 10x - 10x^3}{(x^2+1)^2}$
 $= \frac{10x}{(x^2+1)^2}$
 (iv) $f(x) = 3x^{\frac{1}{2}} - 4x^{-7}$
 $f'(x) = \frac{3}{2}x^{-\frac{1}{2}} + 28x^{-8}$
 $= \frac{3}{2\sqrt{x}} + \frac{28}{x^8}$

(b) y-int at $f(0)$
 $f(0) = -(0-1)^2(0+2)(0+5)$
 $= -10$
 x-int when $f(x) = 0$
 $0 = -(x-1)^2(x+2)(x+5)$
 $x = 1, x = -2, x = -5$

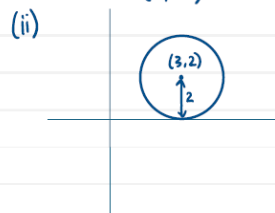
16 (a) (i) $2\sin\theta = -\frac{\sqrt{3}}{2}$
 $\sin\theta = -\frac{\sqrt{3}}{2}$
 $\text{r.a. } \angle = \sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)$
 $= -\frac{\pi}{3}$

 $\theta = \pi + \frac{\pi}{3}, 2\pi - \frac{\pi}{3}$
 $\theta = \frac{4\pi}{3}, \frac{5\pi}{3}$
 (ii) $2\cos^2\theta - \cos\theta = 0$
 $\cos\theta(2\cos\theta - 1) = 0$
 $\cos\theta = 0$ or $2\cos\theta - 1 = 0$
 $\theta = \frac{\pi}{2}, \frac{3\pi}{2}$ $\cos\theta = \frac{1}{2}$
 $\text{r.a. } \angle = \cos^{-1}\left(\frac{1}{2}\right)$
 $= \frac{\pi}{3}$

 $\theta = \frac{\pi}{3}, 2\pi - \frac{\pi}{3}$
 $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$
 $\therefore \theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{3}, \frac{5\pi}{3}$

(b) LHS $= \sec x - \cos x$
 $= \frac{1}{\cos x} - \cos x$
 $= \frac{1}{\cos x} - \frac{\cos x}{1}$
 $= \frac{1 - \cos^2 x}{\cos x}$
 $= \frac{\sin^2 x}{\cos x}$
 $= \sin x \times \frac{\sin x}{\cos x}$
 $= \sin x \tan x$
 $= \text{RHS}$

17 (a)(i) $x^2 - 6x + y^2 + 4y + 9 = 0$
 $x^2 - 6x + y^2 + 4y = -9$
 $x^2 - 6x + 9 + y^2 + 4y + 4 = -9 + 9 + 4$
 $(x-3)^2 + (y+2)^2 = 4$
 $\therefore$ Centre $(3, -2)$ and $r=2$



(b) $y = (2x+1)^4$
 $y' = 4(2x+1)^3(2)$
 $= 8(2x+1)^3$
At $x = -1$
 $y = (2(-1)+1)^4 = 1$
 $y' = 8(2(-1)+1)^3 = -8$
At $(-1, 1) \Rightarrow m = -8$
 $y - 1 = -8(x - (-1))$
 $= -8(x+1)$
 $= -8x - 8$
 $y = -8x - 7$



P(only Quality | reads newspaper)
 $= \frac{15\%}{60\%}$
 $= 25\%$

18 (a)(i) At $t=10$
 $P = 2500e^{-0.036(10)}$
 $= 1744$
(ii) $\frac{dP}{dt} = 2500 \times -0.036e^{-0.036t}$
 $= -90e^{-0.036t}$
At $t=20$
 $\frac{dP}{dt} = -90e^{-0.036(20)}$
 $= -43.81 \dots$
 $\therefore$ declining 44 penguins/year
(iii) When $P=250$
 $250 = 2500e^{-0.036t}$
 $0.1 = e^{-0.036t}$
 $-0.036t = \ln(0.1)$
 $t = \frac{\ln(0.1)}{-0.036}$
 $= 63.96$
 $\therefore$ after 64 years

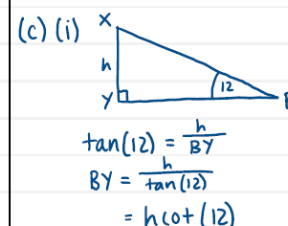
(b) P(at least one wins)
 $= 1 - P(\text{none wins})$
 $= 1 - \left(\frac{8}{9} \times \frac{15}{16}\right)^3$
 $= \frac{91}{216}$

(c) $f(g(x)) = (\sqrt{x-3})^2 + 1$
 $= x - 3 + 1$
 $= x - 2$

$x - 3 \geq 0$
 $x \geq 3$

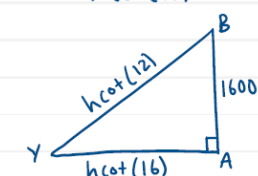
19 (a) $3x^2 - 2mx = -3$
 $3x^2 - 2mx + 3 = 0$
 $\Delta = b^2 - 4ac$
 $= (-2m)^2 - 4(3)(3)$
 $= 4m^2 - 36$
one real solution $\Rightarrow \Delta = 0$
 $0 = 4m^2 - 36$
 $0 = m^2 - 9$
 $m^2 = 9$
 $m = \pm 3$

(b) $\log_4(4x+16) - \log_4(x^2-2) = 1$
 $\log_4\left(\frac{4x+16}{x^2-2}\right) = 1$
 $\frac{4x+16}{x^2-2} = 4^1$
 $4x+16 = 4(x^2-2)$
 $= 4x^2 - 8$
 $4x^2 - 4x - 24 = 0$
 $x^2 - x - 6 = 0$
 $(x-3)(x+2) = 0$
 $x=3$ or $x=-2$



(ii)

$\tan(16) = \frac{h}{AY}$
 $AY = \frac{h}{\tan(16)}$
 $= h \cot(16)$

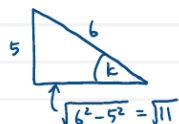


$(h \cot(12))^2 = (h \cot(16))^2 + 1600^2$
 $h^2 \cot^2(12) - h^2 \cot^2(16) = 1600^2$
 $h^2(\cot^2(12) - \cot^2(16)) = 1600^2$
 $h^2 = \frac{1600^2}{\cot^2(12) - \cot^2(16)}$
 $h > 0$
 $\therefore h = \sqrt{\frac{1600^2}{\cot^2(12) - \cot^2(16)}}$
 $= 506.69m$

20 (a) $\sin \theta - \sqrt{3} \cos \theta = 0$
 $\sin \theta = \sqrt{3} \cos \theta$
 $\frac{\sin \theta}{\cos \theta} = \frac{\sqrt{3} \cos \theta}{\cos \theta}$
 $\tan \theta = \sqrt{3}$
r.a.l. $= \tan^{-1}(\sqrt{3})$
 $= 60^\circ$

$\theta = 60^\circ, 180^\circ + 60^\circ$
 $\theta = 60^\circ, 240^\circ$

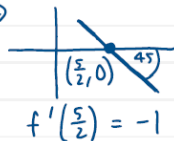
(b) $y = 3 \sin x$
 Goes through $(k, \frac{5}{2})$
 $\Rightarrow$ At $x=k$, $y = \frac{5}{2}$
 $\frac{5}{2} = 3 \sin(k)$
 $\sin(k) = \frac{5}{6}$



However $(k, \frac{5}{2})$ lies in between $\frac{\pi}{2}$ and π ,
 $\therefore$ 2nd quadrant
 $\therefore \tan(k) = -\frac{5}{\sqrt{11}}$

(c) Turning point at $(0,0)$
 $\Rightarrow f(0) = 0$
 $\Rightarrow f'(0) = 0$
 $0 = a + b(0) + c(0)^2 + d(0)^3$
 $a = 0$
 $f'(x) = b + 2cx + 3dx^2$
 $0 = b + 2c(0) + 3d(0)^2$
 $b = 0$

$\therefore y = cx^2 + dx^3$
 Slopes 45° downwards
 at $(\frac{\pi}{2}, 0)$
 $\Rightarrow f(\frac{\pi}{2}) = 0$
 $\Rightarrow$



$f'(\frac{\pi}{2}) = -1$

$0 = c(\frac{\pi}{2})^2 + d(\frac{\pi}{2})^3$
 $0 = \frac{\pi^2}{4}c + \frac{125}{8}d$
 $0 = 50c + 125d$ ①
 $f'(x) = 2cx + 3dx^2$
 $-1 = 2c(\frac{\pi}{2}) + 3d(\frac{\pi}{2})^2$
 $-1 = 5c + \frac{75}{4}d$
 $-4 = 20c + 75d$ ②
 ② $\times 2.5$
 $-10 = 50c + 187.5d$ ③
 ① - ③
 $10 = -62.5d$
 $d = -0.16 = -\frac{4}{25}$
 Sub $d = -\frac{4}{25}$ into ①
 $0 = 50c + 125(-\frac{4}{25})$
 $0 = 50c - 20$
 $c = \frac{2}{5}$
 $\therefore y = \frac{2}{5}x^2 - \frac{4}{25}x^3$
 or
 $y = 0.4x^2 - 0.16x^3$